Who holds government debt?

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Abstract

This paper develops a model in which the composition of public debt across domestic banks, domestic non-banks, and foreign investors is determined by endogenous borrowing limits. The government faces default constraints on non-bank creditors and convex costs of financial repression on bank-held debt. These constraints generate threshold effects: when total debt exceeds a critical level, the government exhausts non-bank borrowing capacity and allocates marginal debt to banks.

We structurally estimate the model using maximum likelihood on panel data of debt composition. The results identify distinct borrowing limits across creditor types and provide a tractable framework to interpret cross-country and time variation in who holds government debt.

Keywords Sovereign Debt; Debt Composition; Financial Repression

1 Introduction

Who holds government debt is central to macroeconomic stability. The distribution of debt across domestic banks, domestic non-banks, and foreign investors shapes default incentives, financial repression, and the transmission of fiscal shocks. Yet most models of sovereign debt focus on the total level of debt and treat its composition as exogenous.

This paper develops a simple framework in which the composition of government debt is endogenously determined by borrowing constraints and repression costs. The government faces default limits on domestic and foreign non-bank creditors, while accumulating debt in the banking sector generates convex financial repression costs. These constraints define a no-default region in debt space and imply threshold effects: beyond a critical level of total debt, the government exhausts non-bank borrowing capacity and reallocates marginal debt to banks.

We structurally estimate the model using panel data on debt composition across countries and over time. The estimation identifies distinct borrowing limits across creditor types and provides a tractable framework to interpret variation in who holds government debt.

1.1 Related literature

The literature on sovereign debt composition and financial repression is extensive. Broner, Martin, and Ventura (2010) study sovereign risk and secondary markets. This section is a placeholder; the full literature review will discuss the determinants of who holds government debt across domestic banks, non-banks, and foreign investors.

2 Model

Decompose total government debt d between components owned by banks (b) and non-banks (n), at home (h) and abroad (f)

$$d = d_n + d_b, \tag{1}$$

$$d_n = d_h + d_f. \tag{2}$$

We assume that the government does not default on home banks, because it is very costly to default on home commercial banks and it does not bring any benefit to default on the home central bank.

The government can default on home non-bank and foreign creditors, d_h and d_f. There are different assumptions one can make regarding default.

Selective default. First, one can assume that the government can default separately on home and foreign creditors.[1] Let us assume that the cost of defaulting on debt of type $\theta = h, f$ is the sum of a fixed cost, γ_θ, plus a variable component proportional to the

[1] Selective default requires market segmentation at the time of default (see Broner, Martin, and Ventura (2010)).

size of debt that is defaulted on, $\lambda_\theta d_\theta$. The benefit of defaulting is $\beta_\theta d_\theta$. The no-default constraint on debt is

$$\beta_\theta d_\theta \leq \gamma_\theta + \lambda_\theta d_\theta, \tag{3}$$

implying an upper bound on the level of debt,

$$d_\theta \leq D_\theta \quad \text{with } D_\theta \equiv \frac{\gamma_\theta}{\beta_\theta - \lambda_\theta}, \quad \theta = h, f. \tag{4}$$

On theoretical grounds, defaulting on residents should be less beneficial than defaulting on foreigners ($\beta_h \leq \beta_f$).[2] Defaulting on residents should be more costly than defaulting on foreigners because it causes more financial disruption at home ($\gamma_h \geq \gamma_f$ and $\lambda_h \geq \lambda_f$). This implies that the government should be able to sell more debt at home, $D_h \geq D_f$.

Condition (4) defines a rectangular region in the (d_h, d_f) space (see Figure 1). For a given total non-homebank government debt d_n, the debt composition can be anywhere on the segment AB. The maximum level of d_n is reached when the $d_n = d_h + d_f$ line goes through the angle C, i.e., $d_h = D_h$ and $d_f = D_f$ and the maximum value of d_n is

$$D_n = D_h + D_f. \tag{5}$$

Comprehensive default. Selective default may be impossible because of market integration or because of *pari passu* clauses in debt contracts. We now show that under some conditions, this does not lead to a very different model of the determination of the composition of debt relative to the case with selective default.

Let us assume that the government either defaults on both d_h and d_f or does not default at all. The benefit of defaulting is $\beta_h d_h + \beta_f d_f$ and the cost of defaulting is $\gamma + \lambda_h d_h + \lambda_f d_f$. Thus, the government does not default if and only if

$$\frac{d_h}{D_h} + \frac{d_f}{D_f} \leq 1, \tag{6}$$

where D_h and D_f are defined as before, setting $\gamma_h = \gamma_f = \gamma$.

Observe that the default region is no longer a rectangle but a triangle (see Figure 2). The maximum level of debt is reached when all the debt is sold at home if $D_h > D_f$. More generally, equation (5) is replaced by $D_n = \max(D_h, D_f)$.

It would make sense, however, to assume that the no-default region is *strictly* convex. For example, if there are coordination problems between home and foreign creditors, the default cost would be higher when those creditors have more equal weights in the holding of debt. The cost of defaulting could be a concave (rather than linear) function of the debt levels d_h and d_f.[3] In reduced form, convexity could be captured by assuming

[2] The sovereign debt literature typically assumes that defaulting on residents has no benefit. However, it could reduce the distortionary taxation required to repay the debt.

[3] This intuition can be formalized by allowing default costs to include an interaction term across creditor groups. Let default costs take the form $C(d_h, d_f) = \gamma + \lambda_h d_h + \lambda_f d_f + f(d_h, d_f)$, with $\partial^2 f / \partial d_h \partial d_f > 0$. Such interaction costs imply higher default costs when debt is held more evenly across home and foreign creditors, generating a strictly convex no-default region as in equation (7).

that (6) is replaced by

$$\left(\frac{d_h}{D_h}\right)^{\rho} + \left(\frac{d_f}{D_f}\right)^{\rho} \leq 1, \tag{7}$$

with $\rho \geq 1$. If there is sufficient convexity, the maximum level of debt D_n is achieved for an interior solution where both residents and foreigners are lending to the government, like in the case of selective default (Figure 2). As ρ goes to infinity we converge to the case with selective default.

What matters for our results is that there is enough convexity in the no-default region to ensure an interior solution. To simplify the analysis, we will assume that this region is a rectangle, like in the case of selective default, but selective default is not a maintained assumption.

Financial repression. The government does not default on banks but accumulating government debt in the banking sector might have costs, for example by crowding out lending to the real sector or because very large bank balance sheets complicate the operation of monetary policy or the management of financial risks.

We assume that the costs of financial repression are zero if bank-held debt d_b is below a threshold D_b. Above that threshold, financial repression has a cost $C(d_b)$ where function $C(\cdot)$ is increasing and convex.

Equilibrium. Assuming no default, for a given level of total debt $d = d_h + d_f + d_b$, the optimal debt composition minimizes $C(d_b)$ conditional on $d_h \leq D_h$ and $d_f \leq D_f$. If the cost of financial repression is very high, the government might prefer to default but we assume this possibility away as our empirical sample does not include any case of default.

The equilibrium is then easy to determine. Let us define the threshold $D \equiv D_n + D_b$. If $d \leq D$, the government can choose any debt composition $(d_\theta)_{\theta=h,f,b}$ such that $d_\theta \leq D_\theta$. If $d > D$, the debt composition is uniquely determined and given by $d_h = D_h$, $d_f = D_f$ and $d_b = d - D_n$.

3 Empirical Analysis of Debt Composition

3.1 Financial Openness and Foreign Private Debt

3.1.1 Measures of Financial Openness

We consider three de jure measures of financial openness—*kaopen*, *cap100*, and *boi_open*—alongside a de facto measure. The *kaopen* index (Chinn–Ito) is based on IMF-reported capital account restrictions and captures the legal openness of a country's financial system, with higher values indicating fewer restrictions. *cap100*, derived from Quinn's coding of capital account regulations, provides a more granular measure of legal openness on a 0–100 scale, where higher values correspond to greater liberalization. *boi_open* is a bond-market-specific measure of openness and is therefore more directly related to foreign participation in government debt markets.

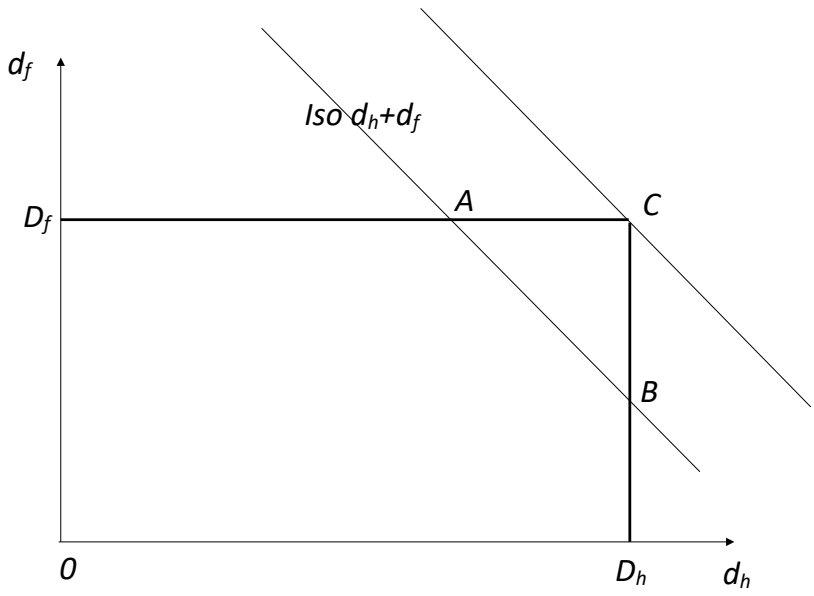

Figure 1 No default region: selective default

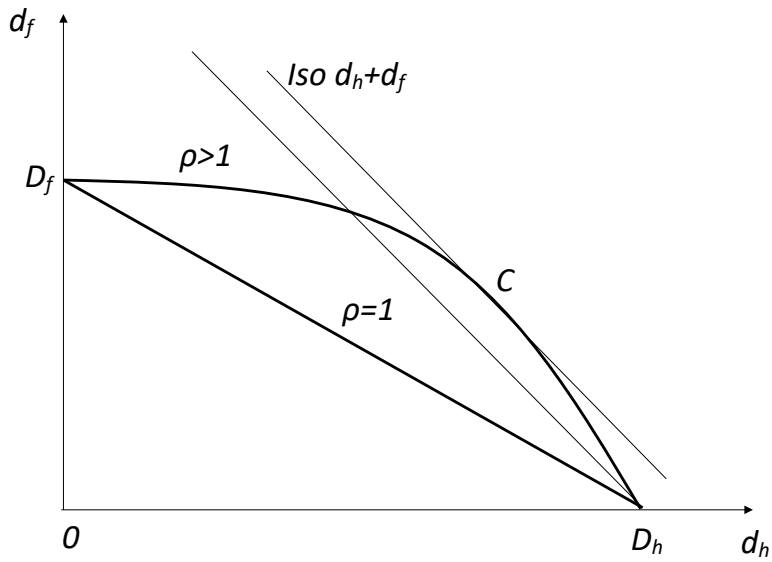

Figure 2 No default region: comprehensive default

Our de facto measure captures realized financial integration through actual cross-border asset and liability positions. Using the External Wealth of Nations (EWN)

dataset, we define

$$\text{De Facto Openness}_{it} = \frac{\text{Total Assets}_{it} + \text{Total Liabilities}_{it}}{\text{GDP}_{it}}.$$

Total assets and liabilities include cross-border positions such as FDI, portfolio equity, and debt, and are measured in current U.S. dollars. Unlike the de jure indices, which reflect policy intent and tend to vary slowly over time, the de facto measure exhibits substantial within-country variation.

3.1.2 Descriptive Evidence on De Jure Openness

Table 1 shows that a large share of countries exhibit no time variation in the de jure openness measures. In particular, 21 out of 27 countries show no variation in *boi_open*, while 15 countries display no variation in either *kaopen* or *cap100*. As a result, fixed-effects estimates using these variables are identified from a relatively small subset of countries.

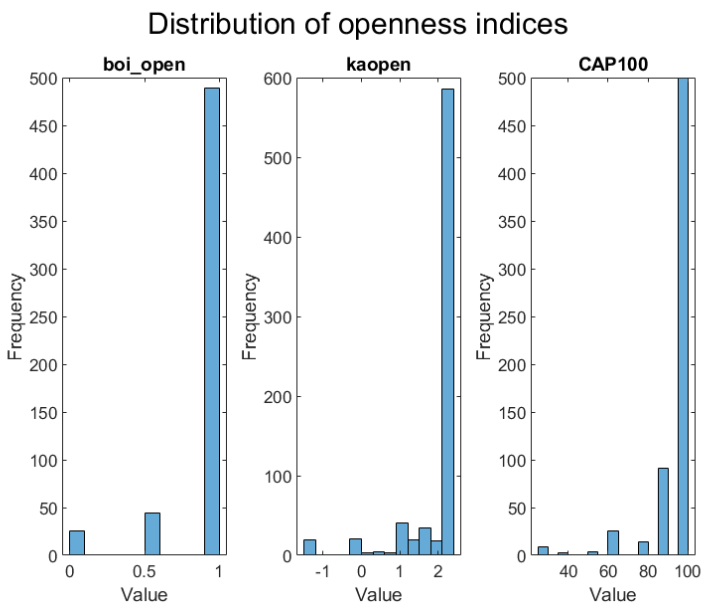

Figure 3 Distribution of de jure openness indices

Table 1 Descriptive statistics for de jure openness measures

Variable	Countries	Mean	Std. Dev.	Between Std.	Within Std.	No variation countries
boi_open	27	0.9141	0.2427	0.1840	0.1604	21
kaopen	31	1.9632	0.7625	0.5836	0.5076	15
cap100	32	94.5433	13.2150	11.4014	8.4157	15

3.1.3 De Jure Openness and the Foreign Share of Government Debt

Table 2 reports the relationship between the foreign share of government debt (*df*) and the three de jure openness measures. In pooled OLS specifications, bond openness (*boi_open*) is negatively and significantly associated with *df*. Once country and year fixed effects are introduced, however, the coefficient on *boi_open* turns positive. This sign reversal suggests that the negative pooled relationship is driven by cross-country differences, whereas within-country increases in bond openness are associated with higher foreign participation in government debt markets.

By contrast, *kaopen* and *cap100* are only weakly related to *df* in pooled OLS, and their fixed-effects coefficients are not statistically significant (not reported). This is consistent with the descriptive evidence above: broad de jure openness measures display limited within-country variation and therefore provide little identifying power in panel regressions.

Table 2 Foreign Debt Share and Financial Integration Measures

	OLS						Fixed Ef
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	df	*df*	*df*	*df*	*df*	*df*	*df*
boi_open	−10.7839***	−6.3342***					4.8903*
	(2.3472)	(2.0997)					(1.727
kaopen			−0.8521	−0.5442			
			(0.9130)	(0.7968)			
*cap*100					−0.1206**	−0.0910**	
					(0.0498)	(0.0436)	
Total debt (*d*)	No	Yes	No	Yes	No	Yes	Yes
Year FE	No	No	No	No	No	No	Yes
Country FE	No	No	No	No	No	No	Yes
Observations	539	539	539	539	479	479	539
R^2	0.0378	0.2537	0.0016	0.2417	0.0121	0.2470	0.393

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Dependent variable is the foreign share of government debt, *df*. *boi_open* is the bond inflow openness measure constructed as $1 - boi$. The fixed-effects column reports the specification for *boi_open* only.

3.1.4 Heterogeneity by Debt Levels

Table 3 examines whether the effect of bond openness differs across debt regimes. For low-debt countries, the effect of *boi_open* on *df* is small and statistically insignificant.

For high-debt countries, by contrast, the interaction term is positive and highly significant. The implied marginal effect is approximately

$$-1.43 + 12.00 \approx 10.6,$$

indicating that bond openness matters much more when governments face greater financing needs.

However, once standard errors are clustered at the country level, the coefficient becomes statistically insignificant, reflecting the limited within-country variation in de jure measures.

Table 3 Bond Openness, Debt Levels, and Foreign Debt

	(1) *df*
boi_open	−1.4266 (1.9019)
High debt (*high_d*)	−15.3740*** (2.1717)
boi_open × *high_d*	12.0008*** (2.4371)
Total debt (*d*)	Yes
Year FE	Yes
Country FE	Yes
Observations	539
R^2 (within)	0.4528

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Dependent variable is the foreign share of government debt, *df*. *boi_open* is the bond inflow openness measure constructed as $1-boi$. High debt (*high_d*) is an indicator for country-years with total debt above the chosen threshold.

3.1.5 De Facto Openness and the Foreign Share of Government Debt

Table 4 reports the relationship between de facto financial openness and the foreign share of government debt. Column (1) shows a negative and significant correlation in pooled OLS. Once total debt (d) is added in column (2), this correlation becomes statistically insignificant. This pattern is driven by countries such as Luxembourg, Cyprus, and Malta, which combine extremely high financial openness with only moderate foreign debt shares.

The sign reverses once country fixed effects are introduced. In column (3), which includes country fixed effects but not year fixed effects, the coefficient on *de_facto* becomes positive and significant. This indicates that the negative pooled relationship is driven by persistent cross-country differences rather than within-country dynamics. In column (4), the coefficient remains positive and significant after additionally controlling for total debt and year fixed effects, suggesting that within countries, greater realized financial openness is associated with higher foreign debt shares over time.

Finally, column (5) interacts de facto openness with a high-debt indicator. The interaction term is not statistically significant, implying that the effect of de facto openness does not differ materially across high- and low-debt regimes. Overall, the results point to a stable positive within-country relationship between realized financial openness and foreign borrowing.

Table 4 De Facto Financial Openness and Foreign Debt

	(1) *df*	(2) *df*	(3) *df*	(4) *df*	(5) *df*
de_facto	−0.0329***	−0.0081	0.0478***	0.0232**	0.0229**
	(0.0094)	(0.0080)	(0.0118)	(0.0093)	(0.0094)
high_d					−5.9187
					(5.0857)
high_d × de_facto					0.0720
					(0.1417)
Total debt (d)	No	Yes	No	Yes	Yes
Year FE	No	No	No	Yes	Yes
Country FE	No	No	Yes	Yes	Yes
Clustered SE	No	No	Yes	Yes	Yes
Observations	747	747	747	747	747
R^2	0.0162	0.3002	0.0188	0.4320	0.4492

Notes: Standard errors in parentheses. Columns (3)–(5) report standard errors clustered at the country level. For the fixed-effects columns, the reported R^2 is the within-R^2. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.2 Domestic Investor Base

3.2.1 Domestic Nonbanks

Table 5 reports the relationship between nonbank financial sector size (*nonbank_size*) and the domestic debt share (*dh*). We define *nonbank_size* as the sum of insurance assets (*di11*) and pension fund assets (*di13*), both measured relative to GDP, from the Global Financial Development (GFDD) dataset. In pooled OLS, *nonbank_size* is positively and significantly associated with *dh*, and this result remains after controlling for total debt in column (2).

Columns (3)–(5) introduce country and year fixed effects. The coefficient becomes smaller and statistically insignificant in column (3), suggesting that part of the pooled relationship reflects cross-country differences. Once total debt is added in column (4), however, the coefficient becomes positive and significant again. The interaction term in column (5) is not statistically significant, indicating no strong evidence that the effect of nonbank size differs across debt regimes.

Table 5 Nonbank Size and Domestic Debt Share (*dh*)

	(1) *dh*	(2) *dh*	(3) *dh*	(4) *dh*	(5) *dh*
nonbank_size	0.1014***	0.0843***	0.0344	0.0485**	0.0439**
	(0.0101)	(0.0077)	(0.0238)	(0.0184)	(0.0194)
high_d					3.6829
					(3.4779)
high_d × nonbank_size					0.0055
					(0.0231)
Total debt (*d*)	No	Yes	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes
Country FE	No	No	Yes	Yes	Yes
Clustered SE	No	No	Yes	Yes	Yes
Observations	547	547	547	547	547
R^2	0.1551	0.5187	0.2268	0.3497	0.3712

Notes: Standard errors in parentheses. Columns (3)–(5) report standard errors clustered at the country level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.2.2 Domestic Private Banks

We use *di02* from the GFDD dataset as a proxy for *bank_ size*. This variable measures bank deposit funding relative to GDP and captures the size of the banking system's deposit base, which in turn reflects banks' balance-sheet capacity to absorb government debt.

Table 6 shows that *bank_ size* is positively associated with the domestic private bank debt share (*dpb*) in pooled OLS. The coefficient becomes smaller once total debt is controlled for, suggesting that part of the baseline relationship reflects the correlation between banking sector size and overall debt levels. In the fixed-effects specifications, the coefficient is weaker and only marginally significant in column (4).

Column (5) shows a positive and significant interaction between *bank_ size* and the high-debt indicator. This suggests that bank size matters primarily in high-debt country-years, where a larger banking sector is associated with a higher share of debt held by domestic private banks.

Table 6 Bank Size and Domestic Private Bank Debt Share (*dpb*)

	(1) *dpb*	(2) *dpb*	(3) *dpb*	(4) *dpb*	(5) *dpb*
bank_size	0.0594***	0.0131*	0.0462	0.0508*	0.0307
	(0.0098)	(0.0070)	(0.0304)	(0.0261)	(0.0185)
high_d					−14.0926**
					(6.5796)
high_d × bank_size					0.1335**
					(0.0583)
Total debt (d)	No	Yes	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes
Country FE	No	No	Yes	Yes	Yes
Clustered SE	No	No	Yes	Yes	Yes
Observations	732	732	732	732	732
R^2	0.0481	0.5421	0.1968	0.3688	0.4350

Notes: Standard errors in parentheses. Columns (3)–(5) report standard errors clustered at the country level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.2.3 Central Banks

Table 7 reports the relationship between central bank size and the central bank share of government debt holdings (*dcb*). The coefficient on *cb_size* is large, positive, and highly significant in every specification. Central bank balance sheets largely consist of government securities, so the relationship is partly mechanical.

Even so, the coefficients remain strong after controlling for total debt and introducing country and year fixed effects. The interaction term in column (5) is positive and statistically significant, indicating that the marginal effect of central bank size on debt holdings is substantially larger in high-debt country-years. These results are best interpreted descriptively rather than causally, given the close accounting link between central bank balance sheets and sovereign debt holdings.

Because the central bank category is closely tied to the central bank's own balance sheet composition, an alternative approach is to treat central bank holdings as a residual category: the share of government debt not held by domestic non-bank investors, private banks, or foreign investors.

4 Parameterizing the model

This section is a placeholder. The full parameterization will describe estimation of borrowing limits D_h, D_f, and D_b on panel data of debt composition.

Table 7 Central Bank Size and Central Bank Debt Holdings (dcb)

	(1) dcb	(2) dcb	(3) dcb	(4) dcb	(5) dcb
cb_size	1.0405***	1.0140***	0.7396***	0.7377***	0.3724**
	(0.0156)	(0.0177)	(0.2006)	(0.2039)	(0.1425)
high_d					−0.9266***
					(0.2697)
high_d × cb_size					0.7294***
					(0.1423)
Total debt (d)	No	Yes	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes
Country FE	No	No	Yes	Yes	Yes
Clustered SE	No	No	Yes	Yes	Yes
Observations	298	298	298	298	298
R^2	0.9373	0.9392	0.6695	0.6698	0.7830

Notes: Standard errors in parentheses. Columns (3)–(5) report standard errors clustered at the country level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5 Comparing fiscal stimulus policies

This section is a placeholder.

6 Robustness in a HANK and SAM Model

This section is a placeholder.

7 Conclusion

This paper proposes a simple model linking sovereign default risk, financial repression, and the composition of government debt. Borrowing limits on non-bank creditors and costs of bank absorption jointly determine how debt is allocated across creditor types. When total debt is sufficiently high, composition becomes constrained and shifts mechanically toward the banking sector.

Appendices

A Details of the HANK and SAM Model

This appendix is a placeholder.

References

Broner, Fernando, Alberto Martin, and Jaume Ventura (2010): "Sovereign Risk and Secondary Markets," *American Economic Review*, 100(4), 1523–1555.